27/5/26

ASSIGNMENT - 1

INDEXING BASICS:-

1) S = "Python"

Print (S[0], S[-1], S[6])

String indexing

char :	P	Y	T	H	O	N
+ (index) :	0	1	2	3	4	5
- (index) :	-6	-5	-4	-3	-2	-1

S[0] → first char → P

S[-1] → last char → n

S[6] → out of range.

o/p :- IndexError : string index out of range.

2) Positive indexing.

Starts from left to right

It starts from 0 at the beginning.

Ex//

S = "Hello"

Print (S[0]) → H

Print (S[1]) → e

Negative indexing.

Starts from right to left

-1 → last character

-2 → second last character.

Ex//

S = "Hello"

Print (S[-1]) → o

Print (S[-2]) → l

3) Indexing

S = "code"

c	o	d	e
0	1	2	3

Print (S[10])

Index 10 does not exist.

Slicing

S = "code"

Print (S[1:10])

o/p :- ode

It take characters from index 1 up to 10. If 10 is not available take until the end.

4)

s = "Mississippi"
0 1 2 3 4 5 6 7 8 9 10

First occurrence of 's' → index 2
Last occurrence of 's' → index 6

5)
s = "abcdef"
-6 -5 -4 -3 -2 -1

Print (s[-1], s[-3], s[-6])

O/P:- f d a

SLICING BASICS:-

6)
Syntax:

s[start : stop : step]

Start → starting index
Stop → Ending index
step → how many positions to move.

7)

s = 'Helloworld"
0 1 2 3 4 5 6 7 8 9

s[0:5] → Hello
s[5:] → World
s[:5] → Hello
s[:] → Helloworld

8) s = "Python"
Print (s[4:1])

In the Positive step the start should be smaller then step so it returns the blank space.

9) By using negative index we can extract the last 3 char of a string using slicing.

Ex:- s = "Programming"

```
Print(s[-3:])
```

O/P:- ing

10) s = "abcdefgh"

Print (s[::1]) → abcdefgh ⇒ Takes every char
Print (s[::2]) → aceg ⇒ Takes every 2nd char from index 0
Print (s[1::2]) → bdfh → Takes every 2nd char from index 1

STEP & REVERSE SLICING:-

11) s = "Python"

```
Print (s[::-1])
```

O/P:- nohtyP

The step value is (-1).

12) s = "abcdefgh"

-8 -7 -6 -5 -4 -3 -2 -1
0 1 2 3 4 5 6 7

Print(s[::-1]) ⇒ hgfedcba ⇒ Reverse full string
Print(s[:: -2]) → hfdb ⇒ Reverse with step -2
Print (s[6:1:-1]) → gfedc ⇒ Start at 6 & stop at 1 reverse order.

13) s = "a1b2c3d4e5"

```
Print (s[1:10:3])
```

O/P:- "12345"

14)
s = "Python"

Print (s[5:0:-1]) → nohty
Print (s [5:: -1]) → nohtyP
Print (s[:0:-1]) → nohty

15)
s="hello"

Print (s[100::-1]) → olleh ⇒ out of range. So automatically last value

Print (s[::-100]) → o

16)
s="abcdefghij"

Print (s[2:-1])

o/p:- cdefghi

17) s=" race car"

Print (s==s[::-1]) ⇒ If reversed string equals orginal string.

o/p: True

18) s=" Hello"

Print (s[1:100]) → ello
Print (s[-100:3]) → Hel
Print (s[100:200])

19)
s=" Hello world"

Print ((s[:6])+"Python")

o/p:- Hello Python

20)
s="abcdefghij"
0 1 2 3 4 5 6 7 8 9

My.slice = slice(1,8,2)
Print (s[My.slice])
o/p:- bdfh

slice (1,8,2) means:

s[1:8:2]